CENTRAL EUROPEAN UNIVERSITY · MSc BUSINESS ANALYTICS · CAPSTONE

Uzbekistan’s Power Sector to 2040

Forecasting demand, bounding uncertainty,
and locating advisory opportunity in a gas-dominated grid

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All data, modelling, analysis and conclusions are the author’s own and were verified against the source notebooks.

A grid that must nearly double

≈78 %
OF GENERATION IS GAS-FIRED

74⟶124 TWh
DEMAND, 2024 → 2040

16–18 %
TRANSMISSION & DISTRIBUTION LOSSES

≈30 yrs
OF SYSTEM DATA ANALYSED

ABOUT THIS BRIEF

This summary distils a capstone study of Uzbekistan's power sector and its path to 2040, prepared for ILF Consulting Engineers Austria. It pairs a probabilistic demand forecast with three policy-calibrated supply scenarios, then locates where engineering and advisory value is most likely to be created.

THE GRID TODAY

Generation	**≈ 78 % gas-fired**
Demand (2024)	**≈ 74 TWh**
T&D losses	**≈ 16–18 %**
Self-sufficiency	**≈ 88 % (2024)**
Data window	**1990 – 2024**

THE OUTLOOK

Electricity demand is set to grow relentlessly. The central forecast lifts consumption from 74 TWh in 2024 to about 124 TWh by 2040 — a 68 % increase — driven by GDP, population and industrial expansion. Even the lower edge of the 90–95 % forecast band keeps climbing, so demand growth is a planning certainty rather than a scenario.

What is uncertain is how that demand is met. Today roughly 78 % of generation is gas-fired, and energy self-sufficiency has slipped from about 125 % in 2010 to 88 % in 2024. Every additional terawatt-hour served from thermal plant therefore tightens the call on domestic gas and imports.

Three futures bound the supply response. They share one demand path and differ only in the ambition of renewable build-out. The investment thesis that emerges is consistent across all three: the near-term prize lies in the grid, in renewable integration and in firm reserve — **not in new baseload generation.**

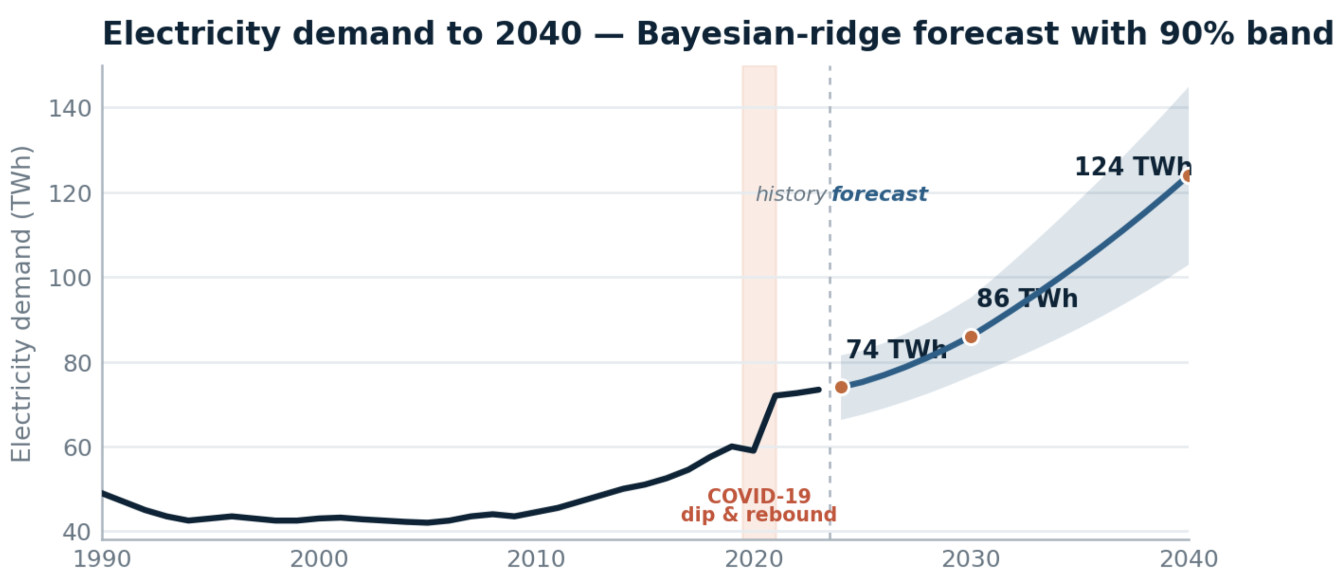

Figure 1 *Electricity demand to 2040 — central path with the 90 % uncertainty band; the 95 % band reaches roughly 103–145 TWh by 2040. Source: notebook 07, advanced forecasting.*

FIVE STRUCTURAL FINDINGS

1
Demand nearly doubles
The central case adds 68 % to reach ~124 TWh by 2040; even the low band rises.

2
The grid is gas-locked
≈78 % thermal today; self-sufficiency has fallen to ~88 %, so more thermal means more imports.

3
Losses are a hidden plant
T&D losses near 16–18 %; closing to ~9 % frees ~7 points of supply — no new generation needed.

4
Uncertainty is widening
The fan opens to roughly ±20 TWh by 2040; the central path implies a reserve gap, not a deficit.

5
Decarbonisation is a choice
2040 renewable share spans 36.5 % to 80 %; carbon intensity 436 vs 90 g/kWh for the same demand.

One demand path, three ambitions

HOW THE SCENARIOS ARE BUILT

All three scenarios run on the **same forecast demand**. Total generation is grossed up from demand by ~9 % to cover network losses; each technology's output is its installed capacity × capacity factor × 8,760 h; and **thermal generation is the residual** that balances the system. The scenarios then differ only in how fast solar, wind and hydro capacity is added — making this a clean test of build-out ambition rather than of demand. *(Source: notebook 07, scenario engine.)*

Business-as-usual DRIFT	Government NATIONAL TARGET	Accelerated TRANSITION
RE SHARE 2040 **36.5 %**	RE SHARE 2040 **60.0 %**	RE SHARE 2040 **80.0 %**
2030 renewables **≈43 %**	2030 renewables **≈58 %**	2030 renewables **≈71 %**
Renewables peak near 43 % by 2030, then slip to 36.5 % by 2040 as thermal plant fills demand growth.	Tracks national policy: ~58 % renewables by 2030, holding ~60 % to 2040. Carbon intensity more than halves.	Front-loads solar and wind to 80 % by 2040 — a near-decarbonised grid at 90 g/kWh.

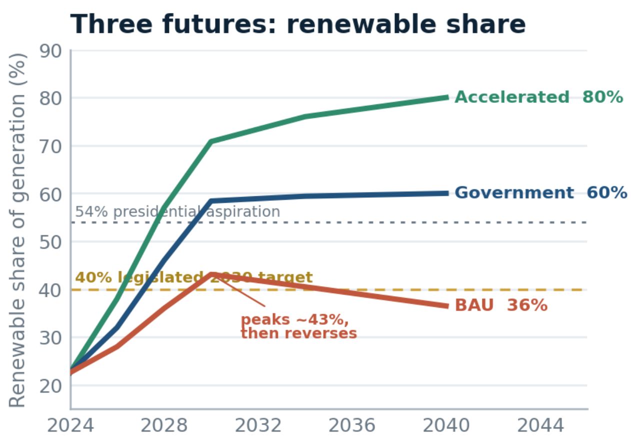

Figure 2 *Renewable share by scenario. BAU peaks then reverses; Government holds 60 %; Accelerated reaches 80 %.*

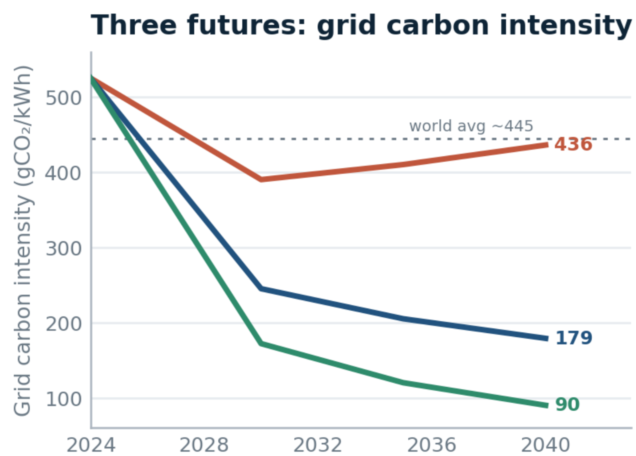

Figure 3 *Power-sector CO_2 to 2040. The gap between paths is the cost of delay — 58.9 vs 12.1 Mt by 2040.*

SCENARIO ENDPOINTS — 2040

METRIC (2040)	BUSINESS-AS-USUAL	GOVERNMENT	ACCELERATED
Renewable share of generation	36.5 %	60.0 %	80.0 %
Thermal generation	85.8 TWh	54.0 TWh	27.0 TWh
Gas burned for power	24.3 bcm	8.9 bcm	4.5 bcm
Carbon intensity	436 g/kWh	179 g/kWh	90 g/kWh
Annual CO_2 emissions	58.9 Mt	24.2 Mt	12.1 Mt
Cumulative capex to 2040	$27.3 bn	$56.9 bn	$79.3 bn

All figures generated by the scenario engine in notebook 07 and the investment-signal workbook (notebook 08).

WHAT THE CHOICE COMES DOWN TO Because all three futures serve the same demand, the only real variable is the pace of renewable build-out — and small differences in that pace compound over time. The paths sit close together to 2030, then fan apart: by 2040 they span 36.5 % to 80 % renewables, 436 to 90 g/kWh of carbon intensity, and 58.9 to 12.1 Mt of CO_2 for an identical quantity of electricity delivered. Reaching the Government target roughly doubles cumulative capital against business-as-usual ($27 bn → $57 bn) yet more than halves carbon intensity, while the Accelerated path adds a further ~$22 bn to secure the last twenty points of renewable share. Crucially, the decisions that fix which path Uzbekistan follows are made in the next few years — which is what makes the investment signals on the following page time-sensitive rather than aspirational.

Where the value is, and where it comes from

THE MODERNISATION PRIZE

≈ 7 points of delivered energy are lost in the network today. T&D losses have sat between 13 % and 18 % for two decades, reaching 17.8 % in 2023. Closing them toward ~9 % would return energy equal to the output of a large new power station — recovered without building any new generation.

Source: notebook 08, T&D-loss signal (investment_signal_td.csv).

TWO FURTHER SIGNALS

1. Reserve headroom, not a deficit

On the central plan the point deficit is zero — but the upper-80 % demand band runs 6.5 TWh above plan in 2024, widening to 14.8 TWh by 2040. The signal is to size firm reserve and flexibility to the band, not the central line.

Source: notebook 08, deficit signal (investment_signal_deficit.csv).

2. Gas-import exposure rises with thermal reliance

Gas burned for power reaches 24.3 / 8.9 / 4.5 bcm by 2040 across the three scenarios. With self-sufficiency already ~88 %, business-as-usual deepens import dependence, while the Accelerated path nearly removes the power sector's gas call altogether.

Source: notebook 08 gas series; supply-driver EDA (notebook 03).

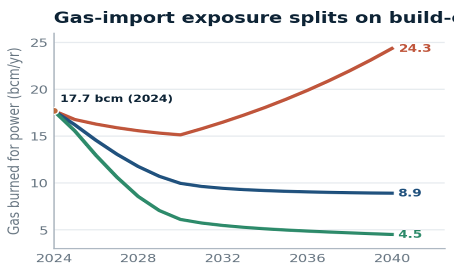

Figure 4 *Gas burned for power by scenario, 2024–2040 — the import-exposure signal.*

WHAT IT COSTS TO BUILD

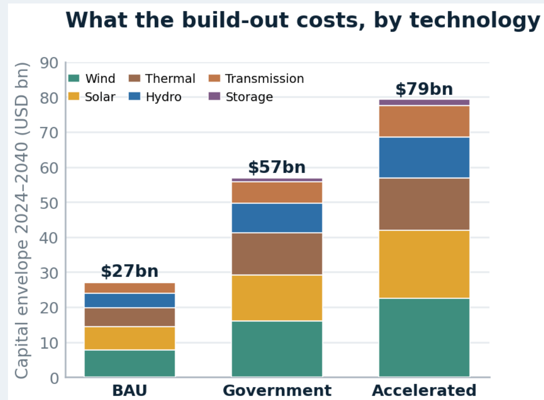

Figure 5 *Cumulative capex 2024–2040 by technology and scenario. Source: notebook 08, investment signals.*

The Government target implies about **$57 bn** to 2040; the Accelerated path roughly **$79 bn**. The largest swings are solar and wind — this is the pipeline against which advisory mandates are scoped.

ADVISORY PRIORITIES FOR ILF

PRIORITY	INTERVENTION	INDIC. CAPEX	HORIZON	ILF ROLE
Generation build-out	≈25 GW new renewables to hit the Government target	**$28.4 bn**	2024–30	Owner's engineer, due diligence, design
Grid modernisation	≈6,000 km HV + 1,450 km wind corridor; cut losses	**$3.0 bn**	Continuous	Network planning, substation design
Renewable integration	≈5 GWh battery storage for ≥40 % renewables	**$2.0 bn**	2025–30	BESS sizing, grid-code compliance
Energy efficiency	Tariff reform + demand-side efficiency (NEEA)	**$1.5 bn**	2024–30	Auditing, M&V, EPC support
PPP / IFI advisory	EBRD / ADB / IFC pipeline (ACWA, Masdar, EDF)	**Fee-based**	Continuous	Lender's engineer, transaction advisory

Priorities and indicative capex from the advisory map in notebook 08.

PLAN B — THE NUCLEAR HEDGE

≈ 72 % **zero-carbon by 2040**

in the Government scenario, once the planned ~2.1 GW Jizzakh nuclear is added.

Outside the three core scenarios, the Jizzakh project — a large light-water reactor plus RITM-200N small modular units, the first in Central Asia — adds firm low-carbon baseload. Modelled across 1.2–3.6 GW, it lifts the Government 2040 clean share from 60 % toward 80 %: a hedge that gains value precisely if renewables underperform.

Source: notebook 08, Plan B nuclear sensitivity.

METHODOLOGY

Demand is modelled with statistical baselines and machine-learning regressors; the Bayesian-ridge model was selected (≈3 % in-sample MAPE, ≈6 % on the pooled four-country validation). The forecast fan comes from bootstrapped residuals, with the three scenarios applied to the central demand path. Drivers and investment signals are developed across notebooks 02–08.

AI declaration

This executive summary was prepared with generative-AI assistance (Anthropic Claude, via Claude Code) for prose drafting, chart styling and document typesetting. All data, modelling, analysis and conclusions are the author's own and were verified against the source notebooks.